

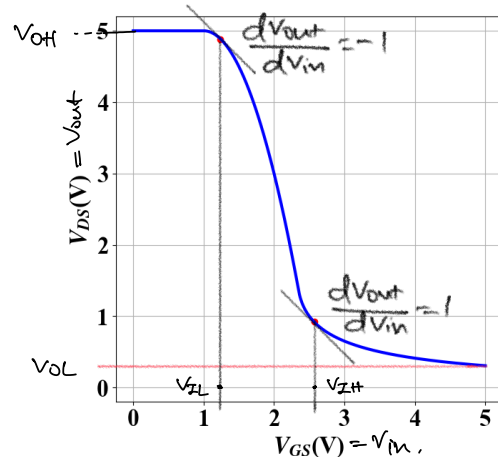
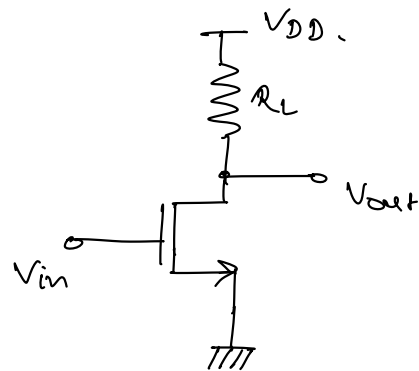
Resistive loaded NMOS Inverter

- $V_{OH} = V_{DD}$
- $V_{OL} = V_{DD} - V_{TO,n} + \frac{1}{k_n R_L}$

$$- \sqrt{\left(V_{DD} - V_{TO,n} + \frac{1}{k_n R_L}\right)^2 - \frac{2V_{DD}}{k_n R_L}}$$

- $V_{IL} = V_{TO,n} + \frac{1}{k_n R_L}$

- $V_{IH} = V_{TO,n} + \sqrt{\frac{8}{3} \frac{V_{DD}}{k_n R_L}} - \frac{1}{k_n R_L}$



$$k_n = (\mu_n C_{ox}) (W/L)_n ; k_p = (\mu_p C_{ox}) (W/L)_p$$

CMOS Inverter:-

$$V_{OH} = V_{DD} ; V_{OL} = 0$$

$$V_{IL} = \frac{2V_{out} - V_{DD} - |V_{TO,p}| + k_r V_{TO,n}}{1 + k_r}$$

$$\text{where, } k_r = \frac{k_n}{k_p}$$

$$V_{IH} = \frac{V_{DD} - |V_{TO,p}| + k_r (2V_{out} + V_{TO,n})}{1 + k_r}$$

Switching Threshold voltage $V_{th} = V_{in} = V_{out}$

$$V_{th} = \frac{V_{TO,n} + \sqrt{\frac{1}{k_r}} (V_{DD} - |V_{TO,p}|)}{\left(1 + \sqrt{\frac{1}{k_r}}\right)}$$

Noise Margin

$$NML = V_{IL} - V_{OL}$$

$$NMH = V_{OH} - V_{IH}$$

